CASE STUDY 1

Power distribution network load flow (linear approximation)

Introduction:

Numerical methods play an important role in solving complex engineering problems that cannot be solved analytically. In electrical distributor network analysis, systems of linear algebraic equations are formed using Kirchhoff's laws. These equations are represented in matrix form and solved using numerical techniques such as the inverse matrix method, Gauss elimination, and iterative methods.
The matrix form provides a systematic and computationally efficient approach for solving large electrical networks using software tools like MATLAB.

$$[A][V] = [B]$$

where:

A = Coefficient Matrix

V = Voltage Matrix

B = Constant Matrix

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Objectives:

1. To analyze a small electrical distributor network using nodal analysis.
2. To represent the system of linear algebraic equations in matrix form.
3. To solve the matrix equation using the inverse matrix method.
4. To compute node voltages and branch currents using numerical techniques.

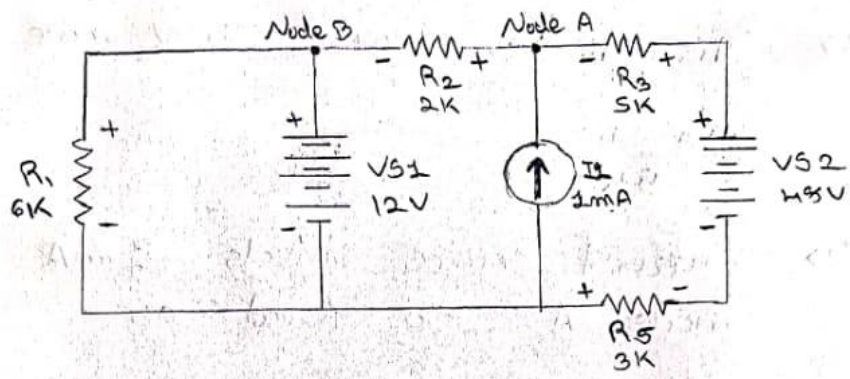
Assumptions:

1. All resistors are linear and obey Ohm's Law.
2. The circuit is operating under steady-state DC conditions.
3. Temperature effects on resistance are neglected.
4. The bottom conductor is considered as the reference (ground) node.
5. Voltage sources are ideal with no internal resistance.
6. Current source is ideal and provides constant current.

Mathematical Modeling:

Power Distribution Network load flow

(Linear Approximation)



-> Step 1 :- Identify Known Values

From the diagram

$R_1 = 6K\Omega$

$R_2 = 2K\Omega$

$R_3 = 5K\Omega$

$R_6 = 3K\Omega$

$I_1 = 1mA$

-> Step 2 :- Node Voltages

Node B is directly connected to a 12V Source to ground

$V_B = 12V$

So we only need to find V_A

-> Step 3 :- Apply KCL at Node A

Currents leaving Node A

1) Through R2 toward Node B

$$\frac{V_A - 12}{2K}$$

2) Through R3 toward 48V node

$$\frac{V_A - 48}{5K}$$

3) Current source injects 1mA into node A so treat as

$-1mA$

→ KCL Equation

$$\frac{V_A - 12}{2000} + \frac{V_A - 48}{5000} - 0.001 = 0$$

→ Step 4 :- Multiply by LCM (10000)

$$5(V_A - 12) + 2(V_A - 48) - 10 = 0$$

Expand :-

$$5V_A - 60 + 2V_A - 96 - 10 = 0$$

$$7V_A - 166 = 0$$

$$V_A = 23.71V$$

→ Final Answers

$$\boxed{V_B = 12V}$$

$$\boxed{V_A = 23.71V}$$

* The current for all branches

$V_B = 12V$

$V_A = 23.71 V$

1) Current through R_1 ($6K\Omega$)

R_1 is between Node B and Ground

$$I_{R1} = \frac{V_B}{R_1} = \frac{12}{6000}$$

$I_{R1} = 2mA$ (Downward)

2) Current through R_2 ($2K\Omega$)

Between Node A and Node B

$$I_{R2} = \frac{V_A - V_B}{R_2} = \frac{23.71 - 12}{2000}$$

$$= \frac{11.71}{2000}$$

$I_{R2} = 5.86 mA$ (A -> B)

3) Current through R_3 ($5K\Omega$)

Between Node A and 48V node

$$I_{R3} = \frac{V_A - 48}{5000} = \frac{23.71 - 48}{5000}$$

$$= \frac{-24.29}{5000}$$

$I_{R3} = -4.86 mA$ (48V node -> A)

Negative means actual direction is

4.86 mA (48V node -> A)

4) Current through R_6 ($3K\Omega$)

- First Find bottom-right node voltage
- Since $V_{S2} = 48V$
- Top = 48V
- Bottom node = $48V - V_{R6}$
- Current through R_3 equals current through V_{S2} equals current through R_6 (series)
- $I_{R6} = 4.86mA$

  Direction :- Right - left (as shown Polarity)

5) Current through V_{S1}

Apply KCL at Node B

Incoming $R_2 = 5.86\ mA$

Outgoing $R_1 = 2\ mA$

$I_{VS1} = 5.86 - 2$

$I_{VS1} = 3.86mA$ (downward)

6) Current Source

Given :-

$I_1 = 1mA$ (upward)

* Final Branch Currents Summary

Branch	Current	Direction
R_1	2 mA	Downward
R_2	5.86 mA	A -> B
R_3	4.86 mA	48 V node -> A
R_6	4.86 mA	Right -> left
V_{S1}	3.86 mA	Downward
Current Source	1 mA	upward

CODE:

```
clc;
clear;

% Resistances (Ohms)
R1 = 6000;
R2 = 2000;
R3 = 5000;

% Known voltages
VB = 12;     % Node B voltage (V)
V48 = 48;    % Right side node voltage (V)

% Current source
I1 = 0.001;  % 1 mA

% Coefficient matrix (from nodal equation)
A = 7;

% RHS value
B = 166;

% Solve for VA using inverse matrix method
VA = inv(A)*B;

% Branch currents

IR1 = VB/R1;           % through R1
IR2 = (VA - VB)/R2;    % through R2
IR3 = (VA - V48)/R3;   % through R3

% Display results
fprintf('Node Voltages:\n');
fprintf('VA = %.2f V\n',VA);
fprintf('VB = %.2f V\n\n',VB);

fprintf('Branch Currents:\n');
fprintf('IR1 = %.4f A (%.2f mA)\n',IR1,IR1*1000);
fprintf('IR2 = %.4f A (%.2f mA)\n',IR2,IR2*1000);
fprintf('IR3 = %.4f A (%.2f mA)\n',IR3,IR3*1000);
fprintf('Current Source I1 = %.4f A (%.2f mA)\n',I1,I1*1000);
```

OUTPUT:

```
Node Voltages:
VA = 23.71 V
VB = 12.00 V

Branch Currents:
IR1 = 0.0020 A (2.00 mA)
IR2 = 0.0059 A (5.86 mA)
IR3 = -0.0049 A (-4.86 mA)
Current Source I1 = 0.0010 A (1.00 mA)
```

Results & Discussion:

Using nodal analysis and linear algebraic equations implemented in **MATLAB**, the node voltages and branch currents of the distributor network were calculated. The bottom conductor of the circuit was taken as the reference (ground).

The computed node voltages are:

- $V_A = 23.71\ V$
- $V_B = 12\ V$

Using these voltages, the currents through the circuit branches were obtained using Ohm's law.

Calculated Branch Currents

Branch	**Current (mA)**	**Direction**
I_{R1}	2.00 mA	From Node B to ground
I_{R2}	5.86 mA	From Node A to Node B
I_{R3}	−4.86 mA	Actual flow from 48 V node to Node A
Current Source I_1	1.00 mA	Upward into Node A

The negative value obtained for the current through R_3indicates that the actual direction of current flow is opposite to the assumed direction during formulation of the nodal equation. This confirms that current flows from the higher potential node (48 V) toward node A.

The results also satisfy **Kirchhoff's Current Law (KCL)** at node A and node B, demonstrating that the algebraic sum of currents at each node is zero. This verifies the correctness of the numerical solution.

The use of matrix equations in MATLAB simplifies solving circuit equations and reduces manual computational effort, especially for larger networks.

Conclusion:

- In this experiment, a small distributor network was analyzed using linear algebraic equations. The system of equations derived from nodal analysis was solved using MATLAB. The node voltages and branch currents were successfully determined.
- The results confirm that numerical methods and computational tools are highly effective for solving electrical network problems. MATLAB provides an efficient platform for solving systems of linear equations, validating circuit laws, and analyzing complex electrical systems with improved accuracy and speed.
- Thus, the study demonstrates the practical application of numerical methods and computational programming in electrical engineering circuit analysis